Table 6.2

$X=x_i$	Number of Household	$p(x_i)$
0	30	0.03
1	320	0.32
2	380	0.38
3	190	0.19
4	80	0.08
Total	1000	1.00

(i) $p(x_i) \geq 0 \ \forall \ i$ and

(ii) $\sum_{i=1}^{\infty} p(x_i)$

$= p(0)+p(1)+p(2)+p(3)+p(4)$

$= 0.03+0.32+0.38+0.19+0.08 = 1$

Hence $p(x_i)$ is a probability mass function.

Note

For $X=0$, the probability 0.03, comes from 30/1000, the other probabilities are estimated similarly.

Example 6.2

A random variable X has the following probability function

Values of X	0	1	2	3	4	5	6	7
$p(x)$	0	a	$2a$	$2a$	$3a$	a^2	$2a^2$	$7a^2+a$

(i) Find a, Evaluate (ii) $P(X<3)$, (iii) $P(X>2)$ and (iv) $P(2<X\leq 5)$.

Solution:

(i) Since the condition of probability mass function is

$$\sum_{i=1}^{\infty} p(x_i) = 1$$

$$\therefore \sum_{i=0}^{7} p(x_i) = 1$$

$$0+a+2a+2a+3a+a^2+2a^2+7a^2+a = 1$$

$$10a^2+9a-1 = 0$$

$$(10a-1)(a+1) = 0$$

$$a = \frac{1}{10} \text{ and } -1$$

Since $p(x)$ cannot be negative, $a = -1$ is not applicable. Hence, $a = \frac{1}{10}$

(ii) $P(X<3)=P(X=0)+P(X=1)+P(X=2)$

$= 0+a+2a$

$=3a$

$= \frac{3}{10} \quad \left(\because \ a=\frac{1}{10}\right)$

(iii) $P(X>2)=1-P(X\leq 2)$

$=1-\left[P(X=0)+P(X=1)+P(X=2)\right]$

$= 1-\frac{3}{10}$

$= \frac{7}{10}$

(iv) $P(2<X\leq 5)=P(X=3)+P(X=4)+P(X=5)$

$= 2a + 3a + a^2$

$= 5a + a^2$

$= \frac{5}{10}+\frac{1}{100} \quad = \frac{51}{100}$

Example 6.3

If $p(x)=\begin{cases} \frac{x}{20}, & x=0,1,2,3,4,5 \\ 0, & otherwise \end{cases}$

Find (i) $P(X<3)$ and (ii) $P(2<X\leq 4)$

Solution:

$P(X<3) = P(X=0)+P(X=1)+P(X=2)$

$= 0+\frac{1}{20}+\frac{2}{20} \quad = \frac{3}{20}$

$P(2<X\leq 4) = P(X=3)+P(X=4)$

$= \frac{3}{20}+\frac{4}{20} \quad = \frac{7}{20}$

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